DATA BREACHES IN COMPANIES: THE 2017 EQUIFAX DATA BREACH CASE STUDY

A CASE STUDY REPORT

submitted by:

JAMILIN GEORGE OLICKAL
(VML24AD062)

to

The APJ Abdul Kalam Technological University

in partial fulfillment of the requirements for the degree of

Bachelor of Technology

In

Artificial Intelligence and Data Science

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ARTIFICIAL INTELLIGENCE AND DATASCIENCE

VIMAL JYOTHI ENGINEERING COLLEGE CHEMPERI

(AUTONOMOUS)

P.O.- 670632, KANNUR, KERALA, INDIA

CONTENTS

SL.NO	TITLE	PAGE NO.
1	Introduction	2
2	Background of the Organization / Case	2
3	Problem or Violation Description	3
4	Analysis (Laws / Principles Violated – Related to GDPR)	3
5	Impact	4
6	Actions Taken or Penalties	5
7	Recommendations / Preventive Measures	6
8	Conclusion	6
9	References	7

LIST OF TABLES

SL. NO.	TABLE NO.	TITLE	PAGE NO.
1	Table 1	Personal Data Compromised in the Breach	3
2	Table 2	Major Consequences for Equifax	5
3	Table 3	Actions Taken After the Breach	5

INTRODUCTION

In the digital age, organizations collect and store enormous amounts of personal and financial information about individuals. This information is used for services such as credit scoring, identity verification, banking, and online transactions. However, storing such large volumes of data also makes organizations a prime target for cybercriminals.

A data breach occurs when unauthorized individuals gain access to confidential or sensitive data. These breaches can expose personal information such as passwords, financial records, identification numbers, and medical records. Data breaches can lead to serious consequences including identity theft, financial fraud, reputational damage, and legal penalties.

One of the most notable cybersecurity incidents occurred in 2017, involving the major credit reporting agency Equifax. The breach exposed sensitive personal data of approximately 147 million individuals, primarily in the United States but also affecting people in other countries.

This case study analyzes the Equifax data breach, examining the causes, security failures, and consequences of the incident. It also evaluates how the breach violated important data protection principles similar to those defined by the General Data Protection Regulation (GDPR). By studying this case, organizations can better understand the importance of implementing strong cybersecurity measures and protecting user data

.

BACKGROUND OF THE ORGANIZATION

Equifax is one of the largest credit reporting agencies in the world. The company plays an important role in the financial system by collecting and analyzing consumer financial data and providing credit reports to banks, lenders, and financial institutions.

The company was founded in 1899 and is headquartered in Atlanta, Georgia, United States. Equifax operates in more than 20 countries and manages vast databases containing sensitive financial information of millions of individuals.

Major Services Provided by Equifax

- Credit reporting services
- Credit scoring for lenders
- Fraud detection and identity verification
- Employment and income verification

- Data analytics for financial institutions

Because of the nature of its business, Equifax stores highly sensitive data such as Social Security numbers, credit histories, and financial records. This makes the company an attractive target for cyber attackers.

PROBLEM DESCRIPTION

The Equifax data breach occurred because of a vulnerability in a web application used by the company. The attackers exploited a security weakness in the Apache Struts software used by Equifax.

Although a security patch had already been released to fix this vulnerability, Equifax failed to update its system in time. As a result, hackers were able to gain unauthorized access to the company's databases.

The breach exposed sensitive personal information including:

- Names of individuals
- Social Security numbers
- Birth dates
- Addresses
- Driver's license numbers
- Credit card details

The attackers had access to the system for several weeks before the breach was discovered. This failure to detect and respond quickly demonstrated weaknesses in the company's cybersecurity practices.

Table 1: Personal Data Compromised in the Breach

Type of Data	Description	Risk
Social Security Numbers	Identity numbers	Identity theft
Birth Dates	Personal details	Fraud risk
Addresses	Residential data	Privacy invasion
Driver's License Numbers	Government ID	Identity fraud
Credit Card Numbers	Financial information	Financial theft

ANALYSIS(LAWS / PRINCIPLES VIOLATED)

Although Equifax is a US-based organization, the incident violated several key principles similar to those outlined in the General Data Protection Regulation (GDPR).

1. Integrity and Confidentiality

GDPR requires organizations to ensure that personal data is protected against unauthorized access, accidental loss, or destruction.

In the Equifax case, sensitive personal data was not adequately protected due to poor cybersecurity practices and delayed software updates.

2. Accountability Principle

Under GDPR, organizations must demonstrate that they have taken appropriate measures to protect personal data.

Equifax failed to maintain proper vulnerability management and security monitoring systems, indicating a lack of accountability in protecting consumer data.

3. Data Protection by Design and by Default

This principle requires organizations to integrate privacy and data protection measures into the design of their systems.

Equifax lacked several important security mechanisms such as:

- Effective encryption practices
- Strong network segmentation
- Advanced intrusion detection systems

4. Transparency and Breach Notification

GDPR requires organizations to notify authorities and affected individuals quickly after discovering a breach.

Equifax delayed public disclosure of the breach, which reduced transparency and damaged consumer trust

IMPACT

The Equifax data breach had serious consequences for individuals, the company, and the global cybersecurity community.

Impact on Individuals

The exposure of personal information created long-term risks for millions of individuals.

Victims faced risks such as:

- Identity theft
- Financial fraud
- Unauthorized credit accounts
- Loss of personal privacy

Since information such as Social Security numbers cannot easily be changed, affected individuals may face risks for many years.

Impact on the Company

The breach caused significant financial and reputational damage to Equifax.

Table 2: Major Consequences for Equifax

Impact	Description
Financial Cost	Over $1.4 billion spent on settlements and security upgrades
Reputation Damage	Loss of customer trust
Legal Issues	Multiple lawsuits and investigations
Stock Price Drop	Significant decline after breach announcement

The incident also led to strong criticism from regulators, governments, and the public.

ACTIONS TAKEN

Following the data breach, several actions were taken against Equifax.

Government agencies conducted investigations to determine the cause of the breach and the company's responsibility. As a result, Equifax agreed to pay approximately **$700 million in settlements and penalties**.

The company also provided free credit monitoring services to affected individuals to reduce the risk of identity theft. In addition, several senior executives resigned after the incident.

Equifax later improved its cybersecurity infrastructure and introduced stronger security measures to prevent future breaches.

Table 3: Actions Taken After the Breach

Action	Description
Investigation	Government and security investigations
Consumer Support	Free credit monitoring services
Security Improvements	Strengthened cybersecurity systems

Legal Settlement	$700 million settlement

PREVENTIVE MEASURES

Organizations can prevent similar data breaches by adopting strong security and ethical practices.

Regular Software Updates

Companies should regularly update their systems and apply security patches to remove vulnerabilities.

Data Encryption

Sensitive personal information should be encrypted so that even if attackers access the data, it cannot be easily misused.

Security Monitoring and Audits

Organizations should conduct frequent security audits and monitoring to detect unusual activities and potential threats.

Employee Awareness and Training

Employees should be trained in cybersecurity practices to reduce human errors and improve overall security.

Compliance with Data Protection Laws

Companies must strictly follow international data protection regulations such as the **General Data Protection Regulation to ensure proper handling of personal data.

CONCLUSION

The 2017 Equifax data breach is one of the most significant cybersecurity incidents in modern history. The breach exposed sensitive personal information of millions of individuals and demonstrated the serious consequences of weak cybersecurity practices.

The incident highlighted the importance of implementing strong data protection policies, maintaining secure systems, and responding quickly to security vulnerabilities.

Organizations that handle large amounts of personal data must prioritize cybersecurity and ensure compliance with international data protection regulations such as the General Data Protection Regulation (GDPR).

By learning from incidents like the Equifax breach, organizations can improve their security practices and better protect consumer data in the future

REFERENCES

1. Equifax Official Reports on the 2017 Data Breach
2. European Union – General Data Protection Regulation Documentation
3. Cybersecurity research articles on corporate data breaches
4. News reports and academic studies related to the Equifax data breach